C4 Energetics – Knowledge Organiser

AQA A-level Chemistry • Set 1: Key Definitions **Papers 1 & 2**

#	Question	Answer
1	Enthalpy change (ΔH)	Heat energy change at constant pressure
2	Sign of ΔH for exothermic reactions	Negative
3	Sign of ΔH for endothermic reactions	Positive
4	Standard conditions	100 kPa and 298 K
5	Standard enthalpy of combustion ($\Delta_c H^\ominus$)	Enthalpy change when 1 mol burns completely in oxygen, under standard conditions, all in standard states
6	Standard enthalpy of formation ($\Delta_f H^\ominus$)	Enthalpy change when 1 mol of a compound forms from its elements under standard conditions, with **elements and product in standard states**
7	$\Delta_f H^\ominus$ of an element in its standard state	0
8	Mean bond enthalpy	Energy to break 1 mol of a bond in gaseous molecules, averaged over many compounds
9	Bond breaking: endothermic or exothermic	Endothermic
10	Bond making: endothermic or exothermic	Exothermic

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AQA A-level Chemistry • Set 2: Equations **Papers 1 & 2**

#	Question	Answer
1	Calorimetry equation	$q = mc\Delta T$
2	Units of q in $q = mc\Delta T$	J
3	Specific heat capacity of water	4.18 J g^{-1} K^{-1}
4	Equation for ΔH from q	$\Delta H = -q \div n$
5	Units of ΔH	kJ mol^{-1}
6	ΔH from bond enthalpies	Σ bonds broken − Σ bonds made
7	ΔH from enthalpies of formation	Σ $\Delta_f H$ products − Σ $\Delta_f H$ reactants
8	ΔH from enthalpies of combustion	Σ $\Delta_c H$ reactants − Σ $\Delta_c H$ products
9	Hess's law	Enthalpy change is independent of the route taken
10	Why mean bond enthalpy values are less accurate	Averaged over different compounds

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AQA A-level Chemistry • Set 3: Calorimetry and Standard States **Papers 1 & 2**

#	Question	Answer
1	Main source of error in calorimetry	Heat loss to surroundings
2	Other error in combustion experiments	Incomplete combustion
3	What m is in $q = mc\Delta T$	Mass of water or solution heated
4	Mass of 1 cm^3 of aqueous solution	1 g
5	Convert J to kJ	÷ 1000
6	Symbol for standard conditions	$^{\ominus}$
7	Standard state of carbon	C(s, graphite)
8	Standard state of oxygen	$O_2(g)$
9	Standard state of bromine	$Br_2(l)$
10	Standard state of water	$H_2O(l)$